

Electrical Machine- II (BT EE403A)



हरियाणा केंद्रीय विश्वविद्यालय
CENTRAL UNIVERSITY OF HARYANA
NAAC ACCREDITED 'A' GRADE UNIVERSITY

Unit-IV

Synchronous Machine

❖ Synchronous Generator Operation

1. Cylindrical Rotor Machine
2. Salient Pole Rotor Machine

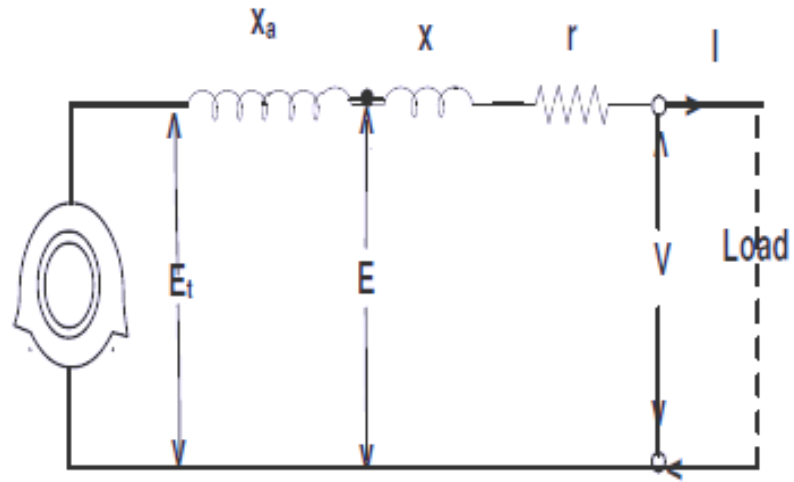
By
Dr. Manish Kumar
Asth. Professor
Department of Electrical Engineering
SOE&T, CUH

Synchronous Generator Operation

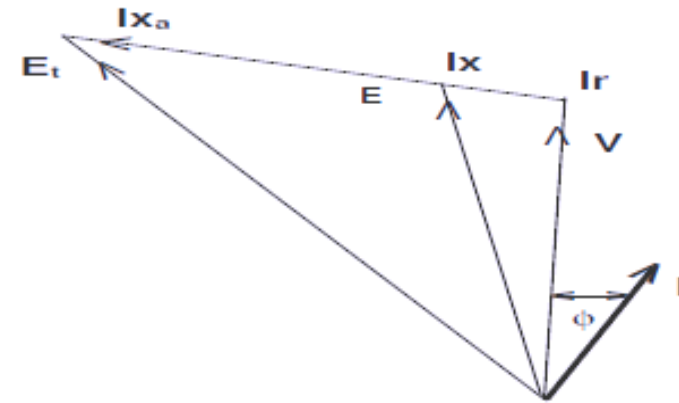
1. Cylindrical Rotor Machine
2. Salient Pole Rotor Machine

1. Cylindrical Rotor Machine

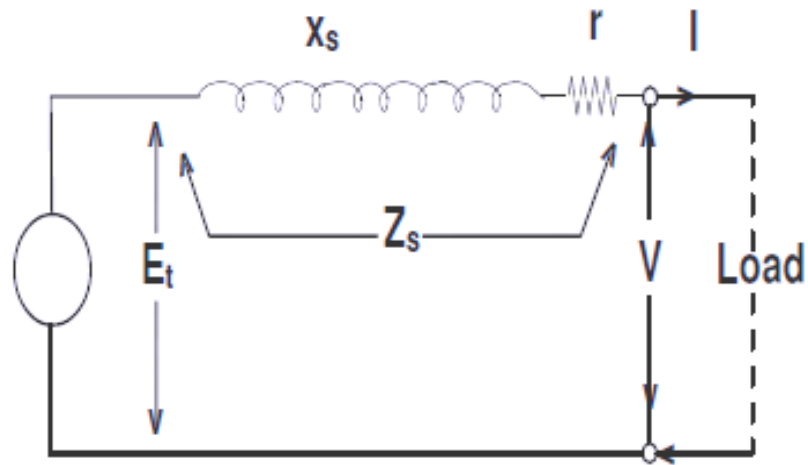
The synchronous generator, under the assumption of constant synchronous reactance, may be considered as representable by an equivalent circuit comprising an ideal winding in which an e.m.f. E_t proportional to the field excitation is developed, the winding being connected to the terminals of the machine through a resistance r and reactance $(X+X_a) = X_s$ all per phase. This is shown in next slide the principal characteristics of the synchronous generator will be obtained qualitatively from this circuit.



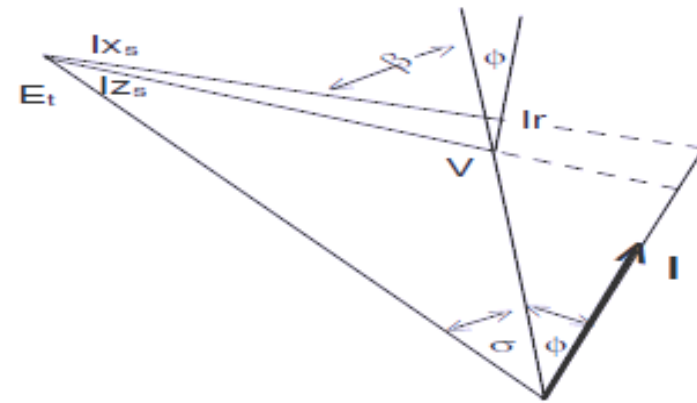
(a)



(b)



(c)



(d)

Equivalent circuits

Generator Load Characteristics

Consider a synchronous generator driven at constant speed and with constant excitation. On open circuit the terminal voltage V is the same as the open circuit e.m.f. E_t . Suppose a unity-power-factor load be connected to the machine. The flow of load current produces a voltage drop I_zs in the synchronous impedance, and terminal voltage V is reduced. The complexor diagram for three values of load. It will be seen that the angle between E_t and V increases with load, indicating a shift of the flux across the pole faces due to cross-magnetization. The terminal voltage is obtained from the complex summation

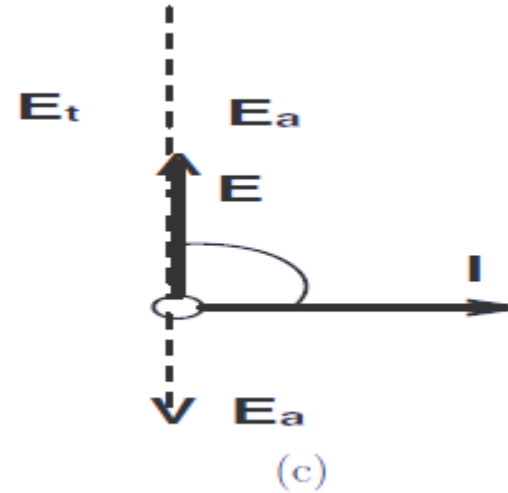
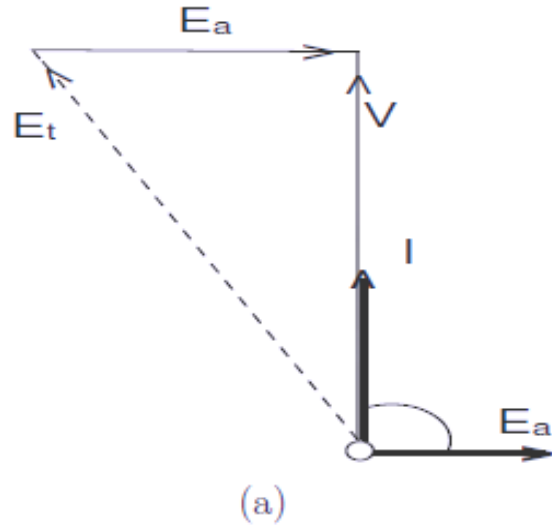
$$V + I_zs = E_t \text{ or } V = E_t - I_zs$$

Algebraically this can be written

$$V = \sqrt{(E_t^2 - I^2X_s^2) - I^2r^2}$$

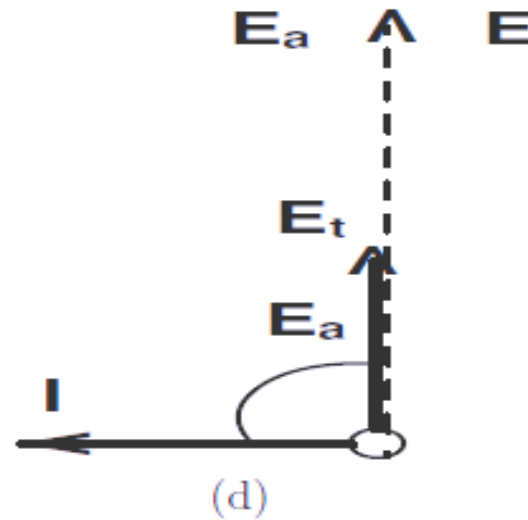
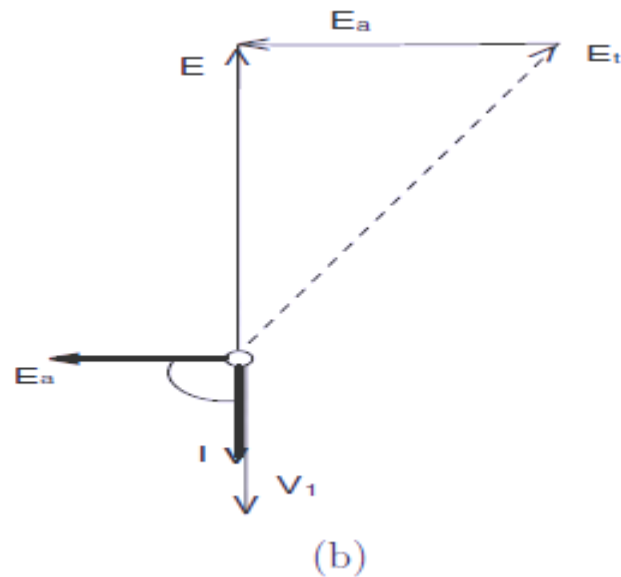
for non-reactive loads. Since normally r is small compared with X_s

$$V^2 + I^2X_s^2 \approx E_t^2 = \text{constant}$$



For a lagging load of zero power-factor, diagram is given in fig. The voltage is given as before and since the resistance in normal machines is small compared with the synchronous reactance, the voltage is given approximately by

$$V \approx E_t - IX_s$$



A leading load of zero power factor will have the voltage

$$V \approx E_t + IX_s$$

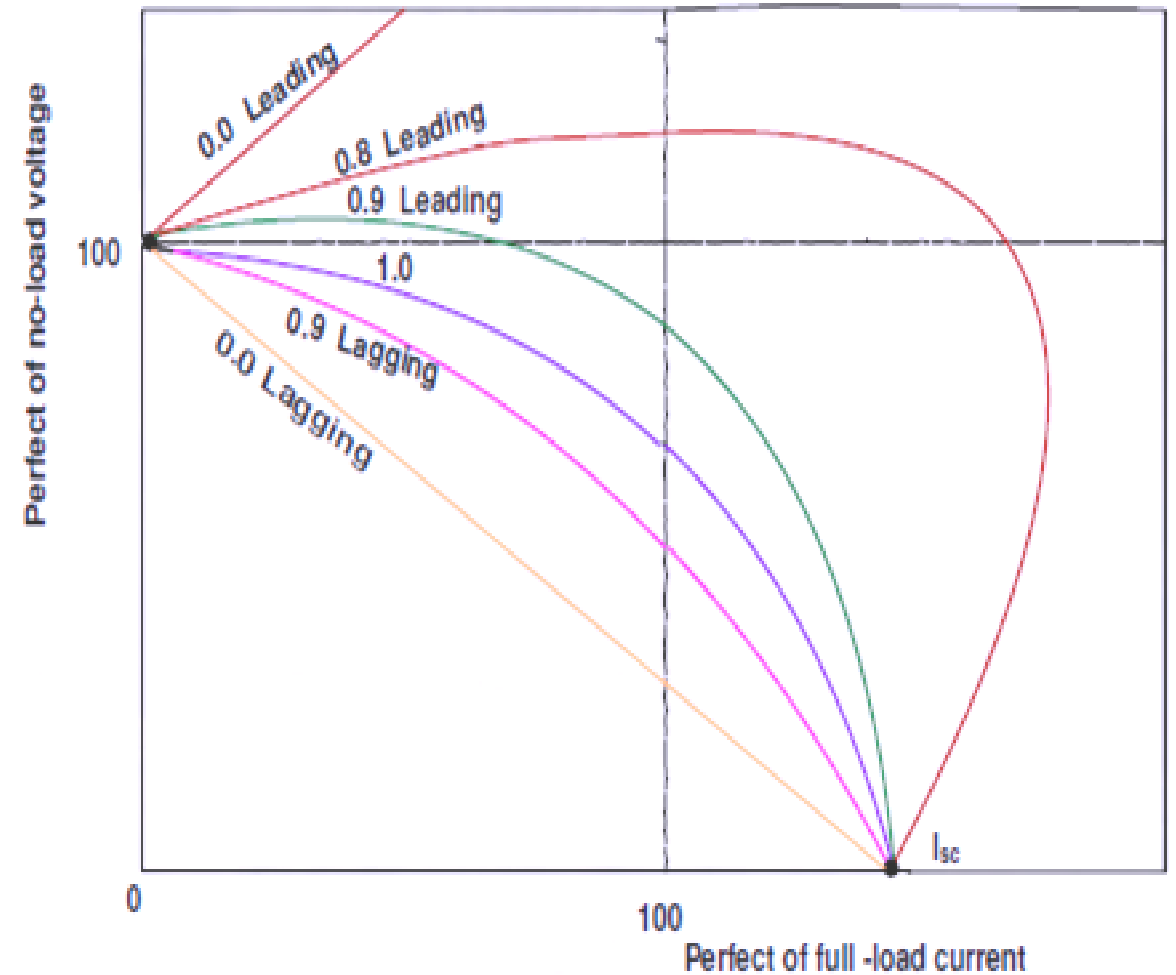
another straight line for which, by reason of the direct magnetizing effect of leading currents, the voltage increases with load.

Variation of voltage with load at constant Excitation

The V/I curve is nearly an ellipse with semi-axes E_t and I_{sc} . The current I_{sc} that flows when the load resistance is reduced to zero. The voltage V falls to zero also and the machine is on short-circuit with $V = 0$ and

$$I = I_{sc} = E_t / Z_s \approx E_t / X_s$$

The voltage-drop with load (i.e. the regulation) is clearly dependent upon the power factor of the load. The short-circuit current I_{sc} which the load terminal voltage falls to zero may be about 150 per cent (1.5 per unit) of normal current in large modern machines.



Generator Load characteristics

Generator Voltage-Regulation

The voltage-regulation of a synchronous generator is the voltage rise at the terminals when a given load is thrown off, the excitation and speed remaining constant. The voltage- rise is clearly the numerical difference between E_t and V , where V is the terminal voltage for a given load and E_t is the open-circuit voltage for the same field excitation. Expressed as a fraction, the regulation is

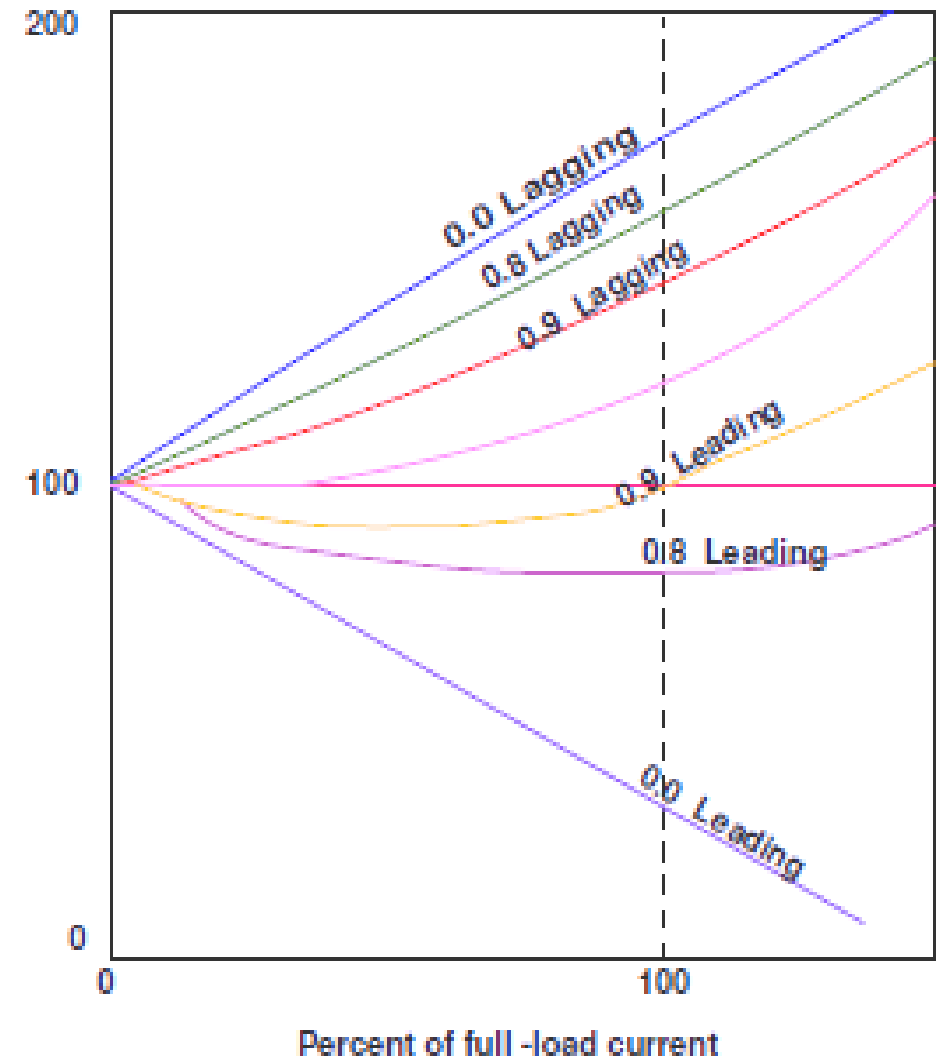
$$\varepsilon = (E_t - V)/V \text{ perunit}$$

Comparing the voltages on full load (1.0 per unit normal current), it will be seen that much depends on the power factor of the load. For unity and lagging power factors there is always a voltage drop with increase of load, but for a certain leading power factor the full-load regulation is zero, i.e. the terminal voltage is the same for both full and no-load conditions. At lower leading power sectors the voltage rises with increase of load, and the regulation is negative. The regulation for a load current I at power factor $\cos \phi$ is obtained from the equality

$$E_t^2 = (V \cos \phi + Ir)^2 + (V \sin \phi + Ix_s)^2$$

Generator Excitation for constant Voltage

- Since the e.m.f. E_t is proportional to the excitation when the synchronous reactance is constant, can be applied directly to obtain the excitation necessary to maintain constant output voltage for all loads.
- All unity-and lagging power-factor loads will require an increase of excitation with increase of load current. Low-leading-power-factor loads, on the other hand, will require the excitation to be reduced on account of the direct magnetizing effect of the zero- power-factor component.
- Typical e.m.f./current curves for a constant output voltage. The ordinates of Fig are marked in percentage of no-load field excitation, to which the e.m.f E_t exactly corresponds when saturation is neglected.



Generator Excitation for constant Voltage

Generator input and output

The electrical power converted from mechanical power input is per phase

$$P_1 = E_t I \cos(\phi + \sigma)$$

Resolving E_t along I

$$P_1 = E_t I \cos(\phi + \sigma) = (V \cos \phi + Ir).I = VI \cos \phi + I^2 R$$

The electrical input is thus the output plus the $I^2 R$ loss, as might be expected. The prime mover must naturally supply also the friction, windage and core losses, which do not appear in the phasor diagram.

In large machines the resistance is small compared with the synchronous reactance so that $\theta = \arctan(x_s/r) \approx 90^\circ$, it can be shown that

$$P = P_1 = E_t I \cos(\phi + \sigma) \approx (E_t/X_s).V \sin \sigma$$

Thus the power developed by a synchronous machine with given values of E_t , V and X_s is proportional to $\sin \sigma$: or, for small angles, to σ , and the displacement angle σ representing the change in relative position between the rotor and resultant pole-axes is proportional to the load power. The term load-, power- or torque-angle may be applied to σ

Salient Pole Rotor Machine

In the case of a salient pole synchronous machine can be taken as two components - one acting along the direct axis (coinciding with the main field pole axis) and the other acting along the quadrature axis (interpolar region or magnetic neutral axis) and as such the mmf components of armature-reaction in a salient-pole machine cannot be considered as acting on the same magnetic circuit. Hence the effect of the armature reaction cannot be taken into account by considering only the synchronous reactance, in the case of a salient pole synchronous machine.

Blondel's two-reaction theory considers the effects of the quadrature and direct-axis components of the armature reaction separately. Neglecting saturation, their different effects are considered by assigning to each an appropriate value of armature-reaction "reactance," respectively x_{ad} and x_{aq} . The effects of armature resistance and true leakage reactance (x_l) may be treated separately, or may be added to the armature reaction coefficients on the assumption that they are the same, for either the direct-axis or quadrature-axis components of the armature current (which is almost true). Thus the combined reactance values can be expressed as :

$$x_{sd} = x_{ad} + x_l \text{ and } x_{sq} = x_{aq} + x_l$$

$$I_{aq} = I_a \cos(\delta + \phi); I_{ad} = I_a \sin(\delta + \phi); \text{ and } I_a = \sqrt{(I_{aq}^2 + I_{ad}^2)}$$

$$I_{aa} = I_a \cos(\phi); I_{ar} = I_a \sin(\phi); \text{ and } I_a = \sqrt{(I_{aa}^2 + I_{ar}^2)}$$

where δ = torque or power angle and ϕ = the p.f. angle of the load.

$$\mathbf{I}_a = \mathbf{I}_{qd} + \mathbf{I}_{aq}$$

The resistance drop $I_a * R_a$ (length AB) is added to the tip of the voltage phasor (OA) in phase with the current phasor (i.e. in a direction parallel to OQ). Then we draw line BC (of length equal to $I_a * x_{sq}$) and extend it to D such that BD will be (of length equal to $I_a * x_{sd}$) at the extremity B of $I_a * R_a$ and at right-angles to I_a . Draw OC and extend it (to F). From D draw the perpendicular DF on OC extended. Then OF represents the induced voltage E . The proof for this can be given as follows:. If DF is extended to G such that this line is perpendicular to BG drawn parallel to OF, we have :

$$BG = BD * \cos(90 - (\delta + \phi)) = I_a * x_{sd} * \sin(\delta + \phi) = I_{ad} * x_{sd} \text{ and}$$

$$GF = CH = BC * \sin(90 - (\delta + \phi)) = I_a * x_{sq} * \cos(\delta + \phi) = I_{aq} * x_{sq}$$

Power relations in a Salient Pole Synchronous Machine

Neglecting the armature winding resistance, the power output of the generator is given by:

$$P = V * I_a * \cos \phi$$

This can be expressed in terms of δ , by noting from Fig. that :

$$I_a * \cos \phi = I_{aq} * \cos \delta + I_{ad} * \sin \delta$$

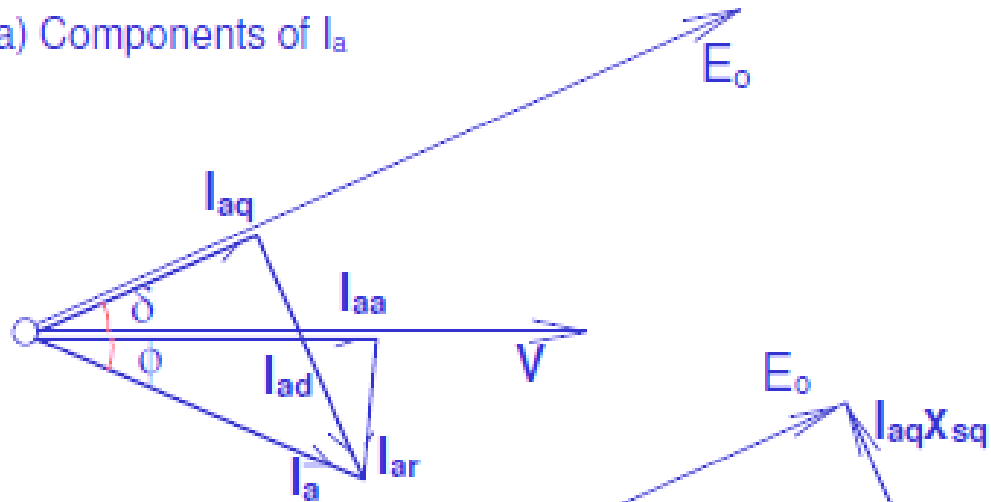
$$V * \cos \delta = E_o - I_{ad} * x_{sd}$$

$$\text{and } V * \sin \delta = I_{aq} * x_{sd}$$

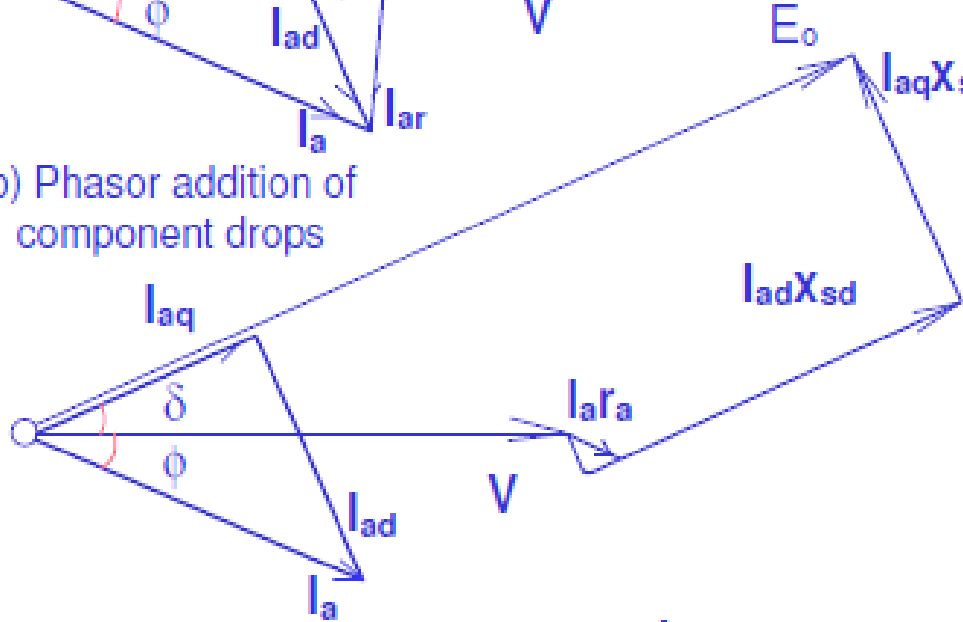
Substituting these in the expression for power, we have.

$$\begin{aligned} P &= V[(V * \sin \delta / x_{sd}) * \cos \delta + (E_o - V * \cos \delta) / x_{sd} * \sin \delta] \\ &= (V * E_o / x_{sd}) * \sin \delta + V^2 * (x_{sd} - x_{sq}) / (2 * x_{sq} * x_{sd}) * \sin 2\delta \end{aligned}$$

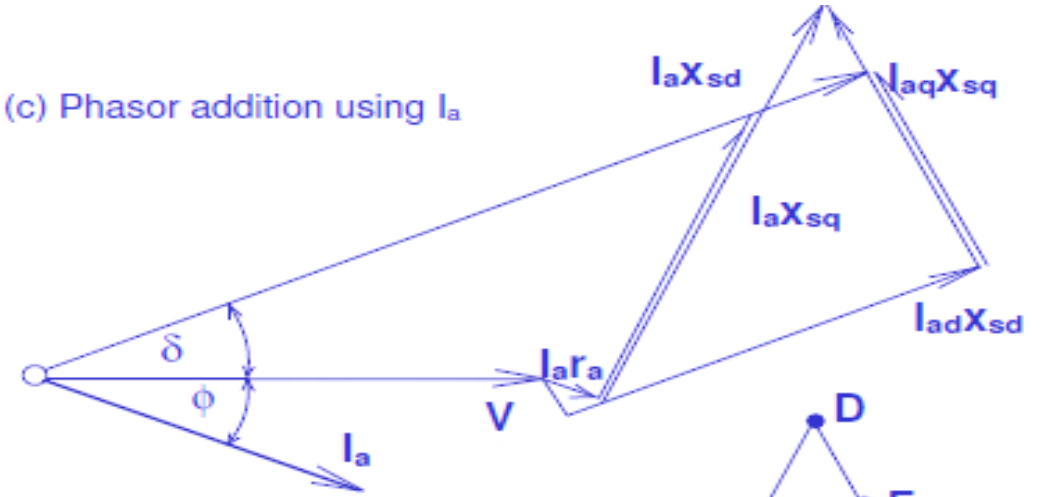
(a) Components of I_a



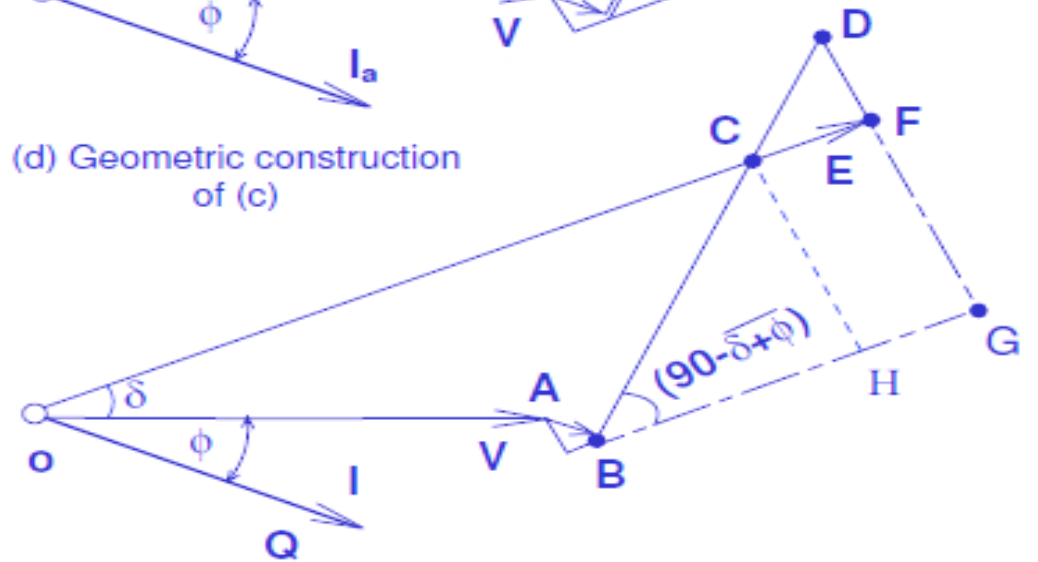
(b) Phasor addition of component drops



(c) Phasor addition using I_a



(d) Geometric construction of (c)



Phasor diagram of a generator-Two reaction theory

It is clear from the above expression that the power is a little more than that for a cylindrical rotor synchronous machine, as the first term alone represents the power for a cylindrical rotor synchronous machine. A term in $(\sin 2\delta)$ is added into the power – power angle characteristic of a non-salient pole synchronous machine. This also shows that it is possible to generate an emf even if the excitation E is zero. However this magnitude is quite less compared with that obtained with a finite E . Likewise we can show that the machine develops a torque - called the reluctance torque - as this torque is developed due to the variation of the reluctance in the magnetic circuit.

Experimental Determination of x_d and x_q

The unsaturated values of x_d and x_q of a 3-Phase synchronous machine can be easily determined experimentally by conducting the following test known as slip test. The rotor of the synchronous machine is driven by means of a prime mover (usually a DC motor in the laboratory) at a speed close to the synchronous speed but not equal to it. The armature is supplied with a low voltage 3-Phase balanced supply through a variac, while the field circuit is kept open. The armature current varies between two limits since it moves through, since the synchronously rotating armature mmf acts through the varying magnetic reluctance paths as it goes from inter-polar axis to pole axis region.

The values of x_{sd} and x_{sq} are determined based on the applied voltage and the armature current values. The ratio of applied voltage to the minimum value of the armature current gives the direct axis synchronous reactance x_{sd} which is usually the same as the synchronous reactance x_s that we usually determine from normal no-load and short-circuit tests. The ratio of applied voltage to the maximum value of the armature current gives the quadrature-axis reactance x_{sq} . For more accurate determination of these values the oscillogram of the armature current and voltage form can be recorded.

Losses and Efficiency

To calculate the efficiency of a synchronous generator, a procedure is to be followed for establishing the total losses when operating under load. For generators these losses are,

1. Rotational losses such as friction and windage.
2. Eddy current and hysteresis losses in the magnetic circuit
3. Copper losses in the armature winding and in the field coils
4. Load loss due to armature leakage flux causing eddy current and hysteresis losses in the armature-surrounding iron.

With regard to the losses, the following comments may be made,

1. The rotational losses, which include friction and windage losses, are constant, since the speed of a synchronous generator is constant. It may be determined from a no-load test.
2. The core loss includes eddy current and hysteresis losses as a result of normal flux density changes. It can be determined by measuring the power input to an auxiliary motor used to drive the generator at no load, with and without the field excited. The difference in power measured constitutes this loss.
3. The armature and field copper losses are obtained as $I_A^2 R_A$ and $V_f I_f$ if Since per phase quantities are dealt with, the armature copper loss for the generator must be multiplied by the number of phases. The field winding loss is as a result of the excitation current owing through the resistance of the field winding.
4. Load loss or stray losses result from eddy currents in the armature conductors and increased core losses due to distorted magnetic fields. ,Although it is possible to separate this loss by tests, in calculating the efficiency, it may be accounted for by taking the effective armature resistance rather than the dc resistance.

After all the foregoing losses have been determined, the efficiency η is calculated as,

$$\eta = \frac{kVAxPF}{kVAxPF + (totallosses)} * 100\%$$

where η = efficiency,

A = load on the generator (output)

PF = power factor of the load

The quantity (kV AxPF) is, of course, the real power delivered to the load (in kW) by the synchronous generator. Thus, it could in general be stated as

$$\eta = \frac{P_{out}}{P_{in}} * 100 = \frac{P_{out}}{P_{out} + P_{losses}} * 100$$

The input power $P_{in} = P_{out} + P_{losses}$ is the power required from the prime mover to drive the loaded generator.